```
CREATE DATABASE IF NOT EXISTS ass5;
USE ass5;

CREATE TABLE Stud_Marks (
   name VARCHAR(100),
   total_marks INT
);

CREATE TABLE Result (
   Roll INT AUTO_INCREMENT PRIMARY KEY,
   Name VARCHAR(100),
   Class VARCHAR(50)
);

DELIMITER //

CREATE FUNCTION Grade_Categorization (marks INT)
RETURNS VARCHAR(50)
DETERMINISTIC
BEGIN
   DECLARE grade VARCHAR(50);

   -- User-defined exception for invalid marks
   DECLARE CONTINUE HANDLER FOR SQLEXCEPTION
      SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Invalid Marks Value';

   -- Validate marks range
   IF marks < 0 OR marks > 1500 THEN
      SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Marks must be between 0 and 1500';
   END IF;

   -- Categorize grade based on marks
   IF marks >= 990 AND marks <= 1500 THEN
      SET grade = 'Distinction';
   ELSEIF marks >= 900 AND marks <= 989 THEN
      SET grade = 'First Class';
   ELSEIF marks >= 825 AND marks <= 899 THEN
      SET grade = 'Higher Second Class';
   ELSE
      SET grade = 'Not Categorized';
   END IF;

   RETURN grade;
END //

DELIMITER ;

DELIMITER //

CREATE PROCEDURE proc_Grade()
BEGIN
   DECLARE done INT DEFAULT 0;
   DECLARE v_name VARCHAR(100);
   DECLARE v_total_marks INT;
   DECLARE v_grade VARCHAR(50);

   -- Declare cursor to iterate through student marks table
```

```
    DECLARE student_cursor CURSOR FOR
        SELECT name, total_marks FROM Stud_Marks;

    -- Handler for end of records
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

    -- Handler for SQL exceptions
    DECLARE CONTINUE HANDLER FOR SQLEXCEPTION
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error occurred during procedure execution';

    -- Open cursor
    OPEN student_cursor;

    -- Loop through each record
    read_loop: LOOP
        FETCH student_cursor INTO v_name, v_total_marks;
        IF done THEN
            LEAVE read_loop;
        END IF;

        -- Call the function for grade categorization
        BEGIN
            DECLARE CONTINUE HANDLER FOR SQLEXCEPTION
                SET v_grade = 'Error in Grade Categorization';
            SET v_grade = Grade_Categorization(v_total_marks);
        END;

        -- Insert results into table
        INSERT INTO Result (Name, Class) VALUES (v_name, v_grade);
    END LOOP;

    -- Close cursor after processing
    CLOSE student_cursor;

    -- Commit changes
    COMMIT;
END //

DELIMITER ;

INSERT INTO Stud_Marks (name, total_marks)
VALUES
    ('Aarav', 1200),
    ('Isha', 950),
    ('Ravi', 880),
    ('Priya', 800);

CALL proc_Grade();

SELECT * FROM Result;
```

-- ----------
-- 💬 VIVA QUESTIONS AND ANSWERS
-- ----------
-- 🔹 1️⃣ What is a Function in MySQL?
-- ➤ A function is a stored program that returns a single value and can be called inside SQL queries.
--
-- 🔹 2️⃣ What is a Stored Procedure?
-- ➤ A stored procedure is a group of SQL statements executed as a block. It can perform actions and doesn't need to return a value.
--
-- 🔹 3️⃣ What is the difference between FUNCTION and PROCEDURE?
-- ➤ Function returns a value and can be used inside SELECT, Procedure performs tasks and uses CALL to execute.
--
-- 🔹 4️⃣ What is a Cursor?
-- ➤ A cursor allows row-by-row processing of query results, used here to loop through all students.
--
-- 🔹 5️⃣ What is DETERMINISTIC in a function?
-- ➤ It means the function always returns the same result for the same input.
--
-- 🔹 6️⃣ What is a HANDLER?
-- ➤ A HANDLER manages exceptions (like NOT FOUND or SQL errors) to prevent abrupt termination.
--
-- 🔹 7️⃣ Why use COMMIT at the end?
-- ➤ COMMIT saves the inserted data permanently after the procedure completes.
--
-- 🔹 8️⃣ Why is Grade_Categorization called inside a procedure?
-- ➤ Because each student's grade is determined dynamically, so we call the function in the loop.
--
-- 🔹 9️⃣ What happens if marks > 1500 or < 0?
-- ➤ A user-defined exception is triggered using SIGNAL, preventing invalid input.
--
-- 🔹 🔟 How can you extend this program?
-- ➤ Add more grade ranges, handle GPA conversion, or store results for multiple semesters.
--
-- 🔹 1️⃣1️⃣ What is SIGNAL SQLSTATE?
-- ➤ It's used to raise custom error messages during procedure or function execution.
--
-- 🔹 1️⃣2️⃣ Why do we need a Cursor here?
-- ➤ To iterate through all rows of Stud_Marks one by one and insert categorized grades accordingly.
--
-- 🔹 1️⃣3️⃣ Real-life application:
-- ➤ Used in universities or institutes to automatically classify student results into grade bands.
--
-- 🔹 1️⃣4️⃣ What does "CONTINUE HANDLER FOR NOT FOUND" do?
-- ➤ It sets the variable "done" when no more rows exist, gracefully ending the loop.

--
-- ◆ 1️⃣ 5️⃣ Can we call this procedure again after adding new data?
-- ➤ Yes, simply insert new records into Stud_Marks and call `CALL proc_Grade();` again.
-- ----------